Coronary heart disease prediction with Machine Learning models

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I. Abstract

This project investigates the use of classification models to predict the risk of cardiovascular diseases (CVDs) using data from the U.S. Behavioral Risk Factor Surveillance System (BRFSS), containing over 300,000 samples. We implemented and compared weighted logistic regression and decision trees, focusing on model interpretability, robustness to imbalance, and generalization. After careful data cleaning and preprocessing, the weighted logistic regression achieved the best performance (F1 = 0.43), highlighting the importance of chronic medical and physical factors as key predictors of CVDs risk.

II. Data Exploration and Processing

A. Exploratory Data Analysis (EDA)

Given the large size of the dataset and the complexity of the problem, we began with an extensive EDA. The goal was to gain confidence with the data, identify patterns, and detect inconsistencies or biases.

We examined single-feature distributions through histograms and boxplots [1, 2], explored feature-target relationships using stratified plots, and used correlation heatmaps and scatterplots to identify redundant or strongly related features. Missing-value patterns and outliers were also inspected to guide subsequent cleaning. This analysis gave us a first idea of which variables are highly correlated with the target and which ones could be omitted without compromising the model performance, while also revealing the presence of special encodings. A cleaning phase was therefore necessary.

B. Data Cleaning

We removed variables with no logical relation to our research scope (e.g., identifiers such as telephone numbers) and those with more than 70% missing values (with a few justifiable exceptions), after verifying the validity of such removal. According to the data description, variables with values composed of 7's and 9's denote missing items and 8's "none". We set these values to NaN and 0 respectively.

C. Feature Grouping and Dimensionality Analysis

Variables were then grouped into broad categories to assess intra-group correlations [3, 4] and missing-value incidence [5]. The categories are: demographics and socioeconomic, health status, chronic conditions, access to care, body measures, physical activity, diet, lifestyle and behaviors. Within each category, redundant features were removed, retaining only the most representative one(s) and with preferably less % of missing values. We computed correlations both within and across categories, as well as between individual variables and the target, obtaining preliminary evidence of the influence of health status on the inspected diagnosis. Inter-category correlations were weak, suggesting relative independence among broader health domains.

To assess dimensionality and redundancy within each feature group, we performed Principal Component Analysis (PCA)[1] on standardized variables and examined the variance explained by the first and cumulative components.

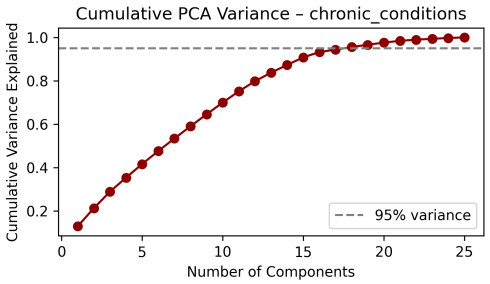


Fig. 1. Variance explained by chronic conditions category components

All categories appeared highly dimensional, with little overlap among components [6]. A preliminary regularized logistic regression was then performed using all retained variables and selected subsets. The comparable results, combined with the PCA findings, led us to keep all 87 variables for the initial training phase, avoiding further reduction to prevent information loss. We later verified this feature set empirically, finding that reducing it further degraded performance.

III. Modeling and Evaluation

Before model training, missing values were imputed using column means, and continuous variables were standardized to zero mean and unit variance. Standardization was applied only for the weighted logistic regression, as decision trees are invariant to feature scaling. We have used F1-Score as our primary metric, as it has a good balance between punishing false positives and false negatives, which is especially important in the medical field. Other considered metrics were ROC-AUC score, precision, recall and accuracy - albeit accuracy to a lesser extent due to the grave class imbalance.

A. Weighted Logistic Regression

We employed a logistic regression model to train and test our predictors. Given the high prevalence of negative outcomes within the dataset, it was necessary to implement a weighted version of the standard logistic regression model to account for this class imbalance.

Model performance was primarily evaluated using the F1 score, combining a 5-fold cross-validation procedure with a grid search, in order to determine the optimal hyperparameters and address potential prediction imbalances.

The search was conducted over various values of λ (the regularization parameter), γ (the gradient descent step size), and

α (the classification threshold for positive diagnoses), while also including the sampling strategy (*none*, *undersampling*, *oversampling*, or *SMOTE*) as a hyperparameter. The results indicated that minimal regularization, relatively large step sizes and and no resampling yielded the highest performance scores, indicating that weighting sufficiently corrected for class imbalance or that balancing methods can lead to other downsides[2].

B. Decision Tree

To further address the underfitting observed in the logistic regression model (see Section IV-A), we implemented a Decision Tree classifier[3]. Unlike linear models, decision trees are not necessarily linear in their coefficients, making it possible to bulid models with higher complexity. This higher level of complexity is expected to be relevant given the size and categorical nature of the dataset. The model used recursive binary splits during training, selecting at each node the split with the highest information gain out of all possible splits. To limit computing time for continuous features, a (randomly selected) cap on possible splits per feature was applied. As in the weighted logistic regression, we employed a 5-fold cross-validated grid search for hyperparameter tuning. The grid parameters included tree depth, minimum samples per leaf, maximum allowed splits per feature, information gain metric (gini-index vs. entropy) and thresholds for the early stopping criterion. This grid search was also combined with the aforementioned technique of oversampling/undersampling and their corresponding ratios.

IV. Results

A. Weighted Logistic Regression

With parameters $\gamma = 0.5$, $\lambda = 0.01$, and $\alpha = 0.8$, the model achieved consistent F1 scores across training and validation folds (average ≈ 0.42) and an F1 of 0.433 on the test set.

These results suggest *underfitting*: performance was similar across training and test data, indicating limited model complexity rather than overfitting. Previous attempts to expand the predictor set brought no major gains, implying sufficient data representation.

Increasing model complexity might improve training results but risks overfitting. To balance this trade-off, a Decision Tree was implemented (see Section III-B). Nonetheless, logistic regression outcomes remain informative: weight variance across folds was minimal, confirming model stability and good—albeit constrained—generalization. This stability also explains why optimal results were obtained with a small λ (weak regularization) and relatively large γ (steeper gradient steps).

The threshold $\alpha = 0.8$ effectively reduced false positives while preserving preventive intent. This aligns with the clinical context, where predictors often reflect gradual physiological changes manageable without undue alarm.

To interpret coefficients, variables were grouped into the same categories used in EDA and aggregated into broader thematic groups [7]. Results highlight body measures as strong predictors, mainly driven by maximal oxygen consumption—consistent with prior evidence.[4]

Among chronic conditions, those contributing most significantly to the predicted diagnosis include obstructive pulmonary diseases[5], diabetes[6], high cholesterol[7], hypertension[8], and a history of stroke. Their medical management exerted the strongest influence on predictions.

Regarding demographic features, sex and age emerged as the most significant predictors. Specifically, being male was associated with a higher likelihood of a positive diagnosis, in line with findings from previous research.[9]

To visualize the relative importance of different feature groups, we plotted the mean absolute logistic regression weights aggregated by combined categories:

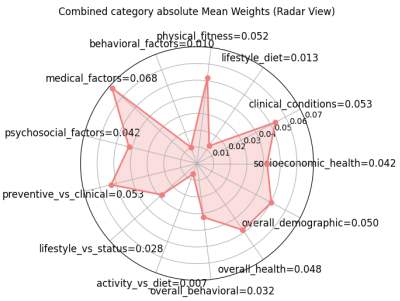


Fig. 2. Mean absolute weights for each combined category

B. Decision Tree

The model was trained using the cleaned and preprocessed data previously described. After running an extensive grid-search cross validation with the parameters described in the previous section, we derived the following results, Since our decision tree does not inherently handle imbalanced data, both oversampling and undersampling performed better than the raw dataset (in terms of F1-score). Oversampling performed more consistently, providing multiple test F1-scores above 0.40. Grid search with CV showed that deeper trees with a high number of data points per leaf performed better, as the ideal parameters turned out to be a depth of 10, with minimum 40 data points per leaf and a surprisingly high sampling ratio of 2.5. We concluded that both the much data per leaf and high sampling ratio prevent overfitting while the depth combated the underfitting.

V. Discussion

In light of the results presented in the previous section we concluded that weighted logistic regression produced the better and more consistent outputs. We validated this result through extensive cross validation as well as by comparing the scores obtained from the grading platform. The similar performance of predictions on training and validation sets suggest a slight degree of underfitting. However, the attempt to combat this phenomenon with more complex solutions such as decision trees, the results became less accurate and more instable.

One of the bigger challenges during this project was the data analysis feature selection. The process of reducing features from more than 300 to just over 80 allows our results to not only be accurate but also easily interpretable. The use of logistic regression further supported our goal to not lose sight of interpretability of our results.

Overall, while even more complex models could potentially lead to more accurate results, we felt like utilizing logistic regression was a good balance of performance and interpretability especially in a highly sensitive field such as medicine.

REFERENCES

1. PCA: Principal component analysis: a review and recent developments
2. See for example: The harm of class imbalance corrections for risk prediction models: illustration and simulation using logistic regression
3. Classic Decision Trees as in Induction of Decision Trees by J. R. Quinlan (1986)
4. See for example Cardiorespiratory Fitness as a Quantitative Predictor of All-Cause Mortality and Cardiovascular Events in Healthy Men and Women.
5. See for example: Cardiovascular risk in chronic obstructive pulmonary disease
6. See for example: Diabetes as a cardiovascular risk factor: An overview of global trends of macro and micro vascular complications
7. See for example: High-density lipoprotein cholesterol and risk of cardiovascular disease
8. See for example High Blood Pressure and Cardiovascular Disease
9. For instance, see Sex differences in risk factors for myocardial infarction: cohort study of UK Biobank participants.